Mechanics of Human Movement

Course Progress 90.2% (11 of 13)

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Assessment

Week 6: Assignment 6

Due date on 2026-09-02, 23:59 IST

The due date for submitting this assignment has passed.Your last recorded submission was on 2026-08-28, 09:44 IST.

1. Internal forces contribute to the acceleration of the system. The statement is — 1 Point

○ True, muscle forces are internal forces and they contribute to acceleration of the system.
○ True, internal forces do not cancel out each other.
● False, internal forces do not appear in the resultant system of forces.
○ False, ground reaction forces contribute to acceleration of centre of mass of human body

Yes, the answer is correct. Score: 1

Accepted answer is False, internal forces do not appear in the resultant system of forces.

2. A standing gymnast raises her arms such that the Centre of Mass (COM) of her body moves upwards by 6cm. The new location of the COM of the arms is higher by — 1 Point

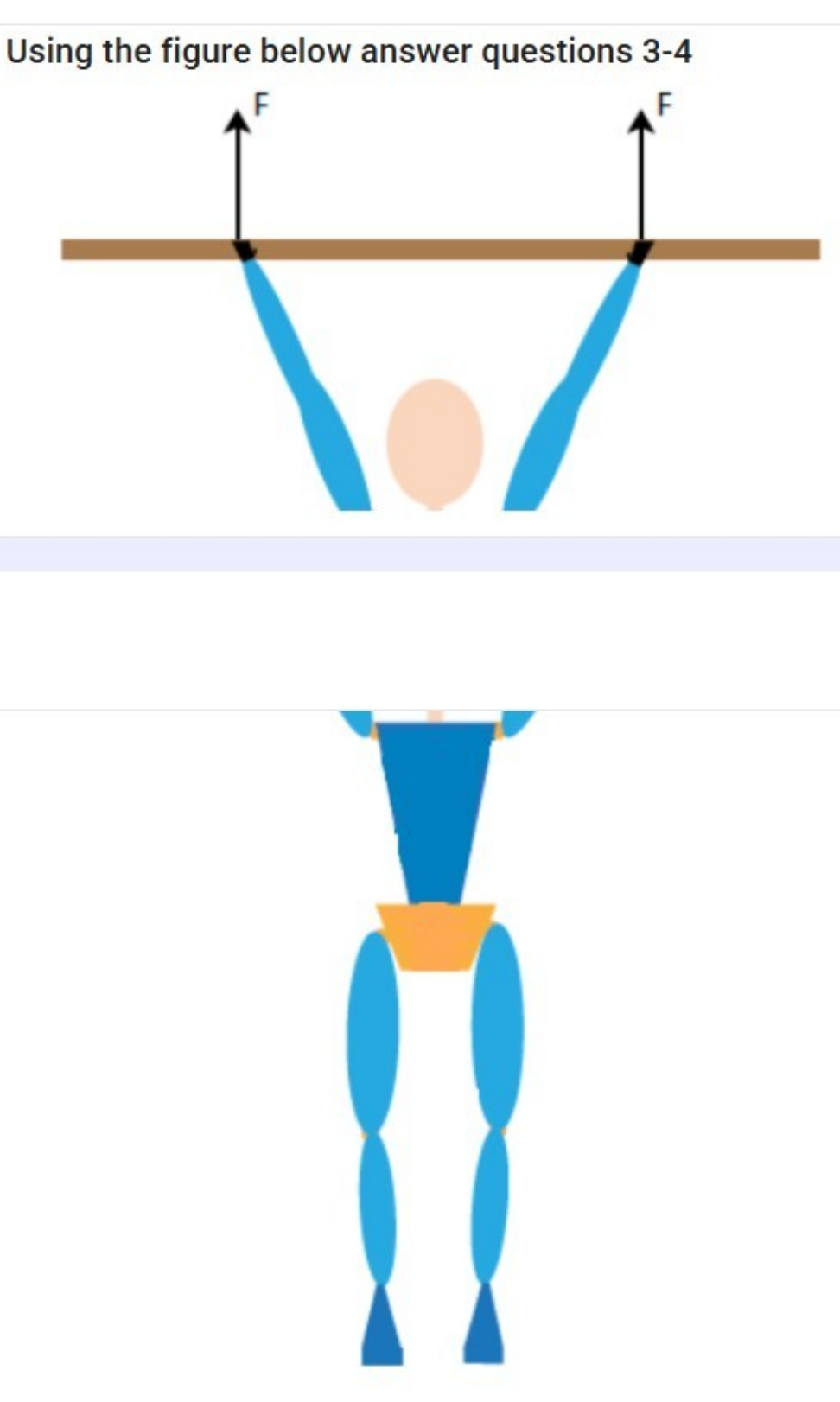
Given:
Mass of the person = 90kg
Mass of each arm = 4kg
Height of the person = 180cm

○ 73.5 cm
● 67.5 cm
○ 6 cm
○ 135 cm

Yes, the answer is correct. Score: 1

Accepted answer is 67.5 cm

Using the figure below answer questions 3-4



3. An athlete is working out her biceps by doing pull ups. During the rising phase, if the athlete's acceleration reaches a peak of $6m/s^2$, the contact force (F) exerted by the bar on the athlete is: — 1 Point

Given: Mass of the athlete is 80kg; $g = 10m/s^2$

○ 160N
○ 517N
● 640N
○ 269N

Yes, the answer is correct. Score: 1

Accepted answer is 640N

4. In solving the previous question, the following is/are considered as internal force(s) to the system. — 1 Point

○ Reaction force exerted on the hand by the bar
○ Reaction force exerted by the bar on the hand
● Muscle forces on the forearm and upper arm
○ Gravitational pull of earth on the athlete

Yes, the answer is correct. Score: 1

Accepted answer is Muscle forces on the forearm and upper arm

5. When we analyse the human body by analyzing the mechanics of the CoM, we are analysing it as a — 1 Point

○ Single rigid body
● Single particle
○ Multiple particles
○ Multi-segmented rigid body

Yes, the answer is correct. Score: 1

Accepted answer is Single particle

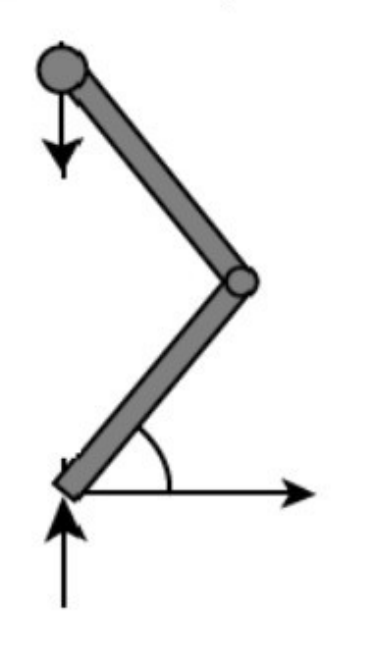
6. Joint reaction forces in multi segment bodies connected through joints are plotted on the free body diagrams using ------- — 1 Point

○ Newton's first law of motion
○ D'Alembert's principle
○ Principle of virtual work
● Newton's third law of motion

Yes, the answer is correct. Score: 1

Accepted answer is Newton's third law of motion

7. Assume the human body is represented as the planar system below with the COM of the upper body mounted on the segments representing the lower limbs. If the foot cannot slide on the ground but can lift off it, the number of degrees of freedom of this system is — 1 Point



○ One
● Two
○ Three
○ Four

No, the answer is incorrect. Score: 0

Accepted answer is Three

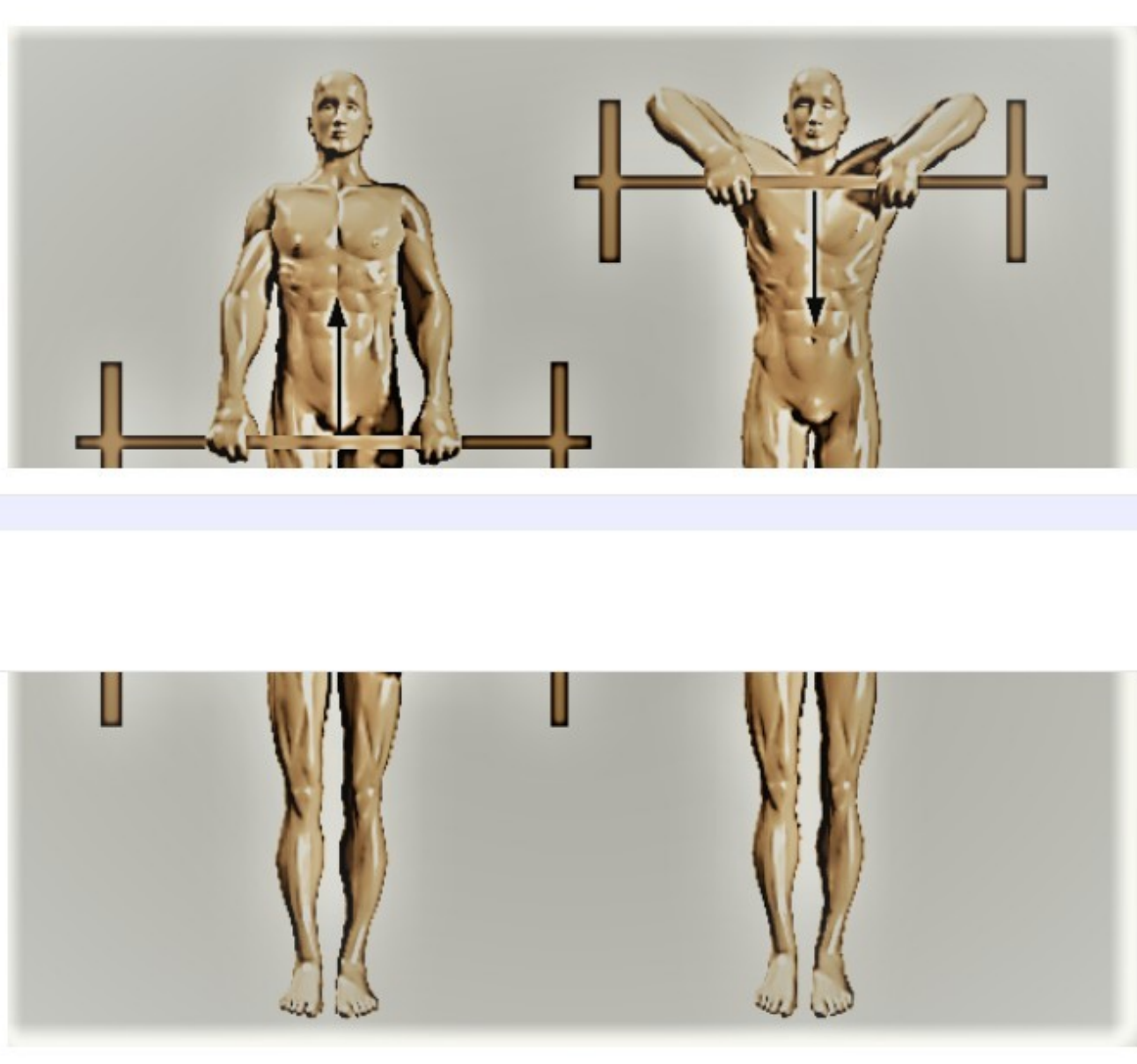
8. When a person is airborne, the person CANNOT — 1 Point

☑ Change the body's centre of mass (COM) by moving their arms
☐ Change the moment of inertia about the COM by moving their arms
☐ Change the acceleration of the COM by moving their arms
☐ Change the angular velocity by moving their arms

Partially Correct. Score: 0.5

Accepted answer is Change the body's centre of mass (COM) by moving their arms, Change the acceleration of the COM by moving their arms

9. — 1 Point



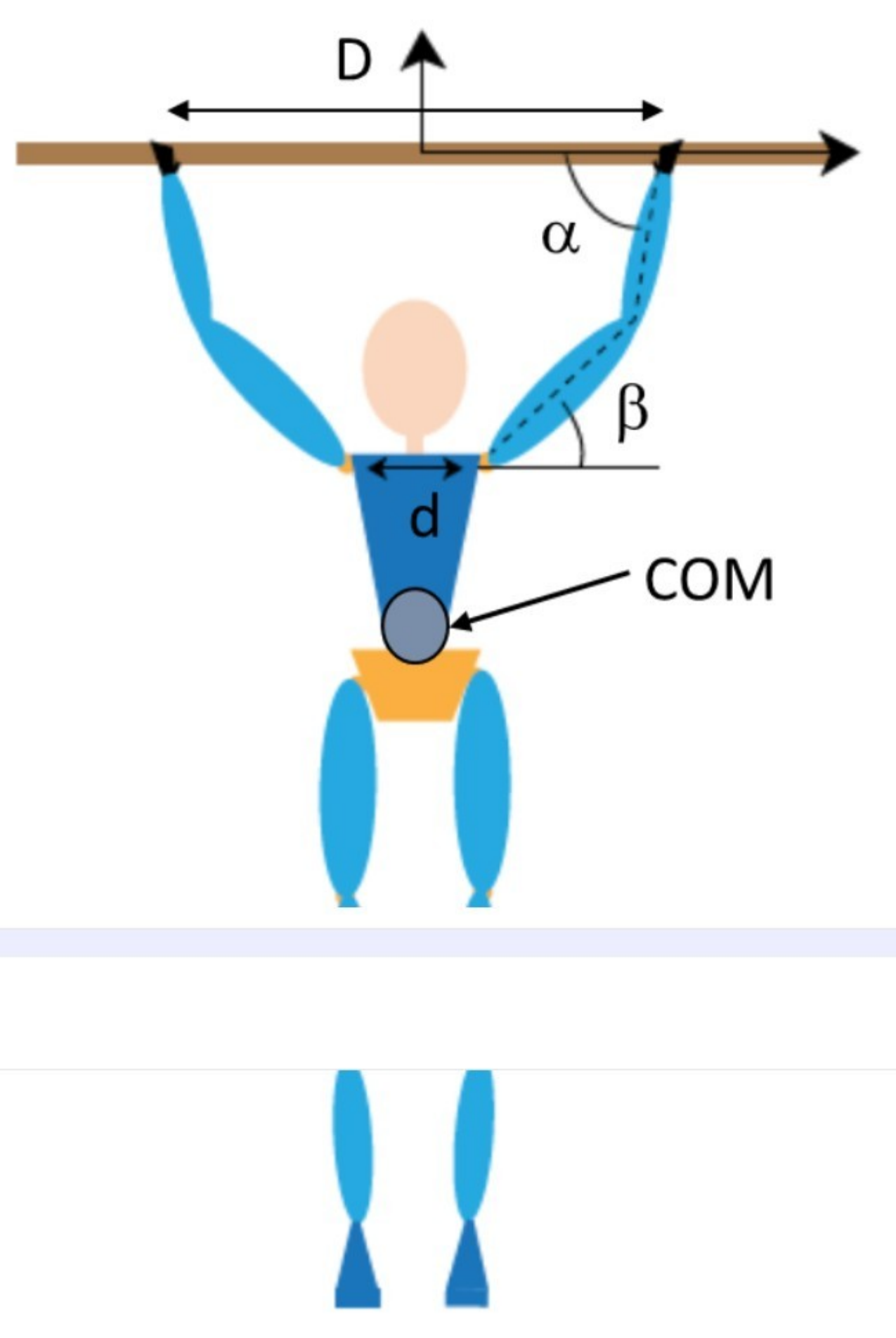
Using a palms down grip and hands close together, an athlete holds a barbell at the thighs at 0.7m above the floor as shown below. Then he pulls the barbell up to his chin with an acceleration of $6m/s^2$ upward. Compute the force the ground exerts on the athlete at the initiation of motion. The athlete is 75kg and the barbell weighs 20kg. Assume the centre of mass of the athlete remains at 0.8m above the floor during the exercise and $g=10m/s^2$.

○ $F_{ground} = 940N$
● $F_{ground} = 1070N$
○ $F_{ground} = 1360N$
○ $F_{ground} = 6700N$

Yes, the answer is correct. Score: 1

Accepted answer is $F_{ground} = 1070N$

A man doing chin-ups by gripping a bar and pulling himself up toward it is modelled as shown in the figure. Let α and β be the angles the forearm and the upper arm, weightless and each of length L, make with the horizontal axis. The COM of the man is located at a distance L below the level of his shoulders. Answer questions 10-12.



10. The relationship between the angles α and β at any instant is given by: — 1 Point

● $D-d=2(L\cos\alpha+L\cos\beta)$
○ $D-d=2(L\sin\alpha+L\sin\beta)$
○ $D-d=(L\sin\alpha+L\sin\beta)$
○ $D-d=(L\cos\alpha+L\cos\beta)$

Yes, the answer is correct. Score: 1

Accepted answer is $D-d=2(L\cos\alpha+L\cos\beta)$

11. At an instant where $\alpha=60°$, $\beta=30°$ and the angular velocity of the upper arm is -0.1 rad/s, the angular velocity of the forearm is — 1 Point

○ 0.6 rad/s
● 0.06 rad/s
○ 0.6L rad/s
○ 0.06L rad/s

Yes, the answer is correct. Score: 1

Accepted answer is 0.06 rad/s

12. At that same instant, the velocity of the COM is — 1 Point

○ 1 m/s
○ L m/s
○ 0.06 m/s
● 0.06L m/s

Yes, the answer is correct. Score: 1

Accepted answer is 0.06L m/s

13. During a triple axel, a skater covers a horizontal distance of 3m and achieves a height of 0.6m. Determine the initial velocity as she lifts off the ice, where V_h and V_v are the velocities of the skater in the horizontal and vertical directions ($g=9.8m/s^2$) — 1 Point

● V_h=4.3m/s V_v=3.4m/s
○ V_h=3.2m/s V_v=7.4m/s
○ V_h=2.5m/s V_v=7.3m/s
○ V_h=2.5m/s V_v=1.9m/s

Yes, the answer is correct. Score: 1

Accepted answer is V_h=4.3m/s V_v=3.4m/s

14. In the mathematical derivation of the angular momentum for a system of particles in space with respect to a point i defined by a Cartesian coordinate system with origin o the following equation is obtained. Which of the following option(s) is/are true? — 1 Point

$$\frac{dH_o}{dt} = \sum_{i=1}^{N} \frac{dr_{i/o}}{dt} \times m_i v_i + \sum_{i=1}^{N} r_{i/o} \times m_i \frac{dv_i}{dt}$$

☑ In the first term, the derivative of the position vector is parallel to the velocity vector, making the first term equal to zero
☐ In the first term, the derivative of position vector is orthogonal to the velocity vector, making the first term equal to zero